Less is More: Early Stopping Rollout for On-Policy Distillation

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Abstract

On-policy distillation has recently emerged as a promising alternative to standard sequence-level imitation, training a student by scoring its own rollouts with a teacher model. However, one important limitation is observed in this paradigm: at each position, the teacher scores the student's next token conditioned on the student's own previous outputs. Early tokens are therefore judged mainly in the context of the original problem and reflect how the student approaches the problem, whereas later tokens are constrained by increasingly student-specific context and provide much weaker supervision. Based on this intuition, we propose Early Stopping Rollout (ESR), a simple yet effective distillation strategy that restricts both rollout generation and reverse-KL to the first N response tokens. This position-focused objective emphasizes the tokens that encode planning and reasoning, rather than spending supervision budget on late tokens whose informational value rapidly degrades. Empirical experiments have proved that ESR shows both surpassing performance with high efficiency under various models and training methods. It is also proved that the teacher collapses in sufficiently long rollouts suggesting that late-position supervision contributes little useful learning signal. We further investigate the mechanism behind this effect and show that position is a distinct and crucial dimension of supervision, not merely a proxy for token uncertainty or difficulty. It establishes position-aware supervision as an important direction for understanding and advancing OPD in future studies.

1 Introduction

On-policy distillation (OPD) has emerged as a dominant paradigm for model distillation in industrial practice. The student generates its own rollouts τ, which are then scored by the teacher: at each token, the teacher's probability $\pi_{teacher}(\tau_{student} \mid x_{prompt})$ serves as the soft target for the student [Agarwal et al., 2024, Gu et al., 2024]. Viewed through an RL lens, this is equivalent to using the teacher as a dense, token-level reward model that judges the student's own behavior on a given prompt. The final objective is then a simple *uniform average* of the per-token reverse-KL across the trajectory.

This uniform averaging implicitly assumes that the teacher's signal is equally reliable at every position in the trajectory. We argue that this assumption is unfounded, and that it breaks down precisely because of the autoregressive structure of LLMs. At the first token, the teacher's score is conditioned only on the prompt: $\pi_{teacher}(\tau^{t=1}_{student} \mid x_{prompt})$ - indeed what we expect the teacher to score on. At a later position m, however, it becomes conditioned also on the student's own previously generated tokens: $\pi_{teacher}(\tau^{t=m}_{student} \mid x_{prompt}, \tau^{t=1:m}_{student})$. This conditioning context is off-policy to the teacher model and may even be out-of-distribution (OOD), and grows more so with position.

To probe how severe this drift is, we run a preliminary experiment that measures how the teacher behaves when continuing from an early-stopped student rollout. As shown in Figure 1, the teacher's quality collapses toward the student's within roughly 300 tokens. If the teacher's late-position signal is unreliable, the natural question is whether early tokens alone are systematically more informative — and whether training on them is sufficient.

Motivated by this, we propose Early Stopping Rollout(ESR): restrict the student rollout to its first N tokens (e.g., 50 or 100) and compute the distillation loss only on this early window. The change is a single line in any on-policy distillation loop. Despite its simplicity, ESRconsistently outperforms full-rollout OPD across tasks (math, code, function calling), training regimes (LoRA, full fine-tuning), model scales (students 1.5B–7B, teachers 1.7B–14B), and model families (Qwen, Gemma), while reducing wall-clock cost by up to $24\times$ and peak training memory by up to $4\times$. We further observe that full-rollout OPD frequently destabilizes and collapses during training, whereas ESRdoes not collapse in any of our settings. Perhaps most surprisingly, although the student is normally upper-bounded by the teacher's performance, ESR-trained students often *exceed* the teacher.

Such a small change yields such large gains probe us to investigate deeper. Firstly, we want to further validate whether the effectiveness of earlier tokens is intrinsically caused by better KL and entropy signals, because we also find that early tokens generally have larger KL and entropy. We test selectors that pick tokens by high KL, high entropy or combined regardless of position; all underperform ESR, with most underperforming full-rollout. Therefore, although we don't exclude KL and entropy as auxillary factors, we confirm that the position is a an independent factor rather a saliency proxy. Furthermore, we identified an additional important mechanism *Convergence Cascade*: after training on the early window, KL divergence on the *untrained* late tokens also drops by 30–40%. Therefore, the KL divergence does not have to see late positions to repair them.

Lastly, we investigated into why ESRsometimes even *exceeds* the teacher — a stronger phenomenon we observe on most of our main experiment Table 1. We trace this to the mode-seeking nature of reverse-KL: the student commits to a subset of supported teacher modes instead of chasing all of them. Our ESRloss helps it steer toward a non-dominant but correct mode rather than the teacher's argmax. Empirically, we find ESR-trained students commit harder to their own choices than full-rollout students yet agree *less* with the teacher's argmax, and produce trajectories $\sim 3\times$ shorter than full-rollout and $\sim 2\times$ shorter than the base student (§5.3). It learns the teacher's strategy but preserves its succinct style, leading to better compount behaviors. This finding showcases a potential path of superceding the teacher model in distillation that worth further investigations.

Our contributions are summarized below:

1. **Early Stopping Rollout(ESR) outperforms full-rollout on-policy distillation.** A one-line change—restricting the rollout length to the first N response tokens—beats full-sequence distillation across tasks, model families, scales, and training regimes, while being dramatically more efficient and far more stable to train. It can even exceed the teacher's own performance (Sections 3–4.3).
2. **A systematic investigation of why it works.** First, through a token-selection ablation, we show that the gain is not "high KL" or "high entropy" in disguise: position itself is a load-bearing axis. We also identify convergence cascading to be another important mechanism. Furthermore, we explain why ESRoften surpasses the teacher: by exploiting the mode-seeking behavior of reverse-KL training, ESRpushes the student toward a better, non-dominant sub-mode (Section 5).

2 Why Late-Position Supervision Is Unreliable

In on-policy distillation, the teacher model scores the student's rollout at each position t via $\pi_t(\cdot \mid x, y_{<t}^{\text{student}})$ and averages these scores uniformly across the rollout. The conditioning context $y_{<t}^{\text{student}}$, however, is whatever this particular student happened to write at this particular moment of training. As t grows it drifts away from any distribution the teacher itself was trained on—off-policy to the teacher and, for long enough or atypical enough rollouts, plausibly out-of-distribution. We term this problem as "Off-policy Teacher Drift".

To verify the intuition, we feed the teacher a k-token student-generated rollout on MATH-500 and letting it continue autoregressively, the teacher's avg@4 (n=4, t=0.7) falls from its 65.30% unconditional baseline to 62.70% at k=100 and 51.75% at k=300—essentially student-baseline

(Figure 1). Late-position teacher scores are not independent assessments of the prompt; it is a guess at how to continue a trajectory the student has already committed to.

The problem compounds during training because the student's policy is moving. Early on, the student's rollout is far from the teacher; late-position targets are fitted under a strongly off-policy prefix. As training progresses and earlier tokens approach the teacher's distribution, the right late-position behavior changes too—continuations that were sensible above an off-policy prefix are no longer sensible above an on-policy one. But the late-position habits learned earlier do not cleanly erase: gradient descent gives no guarantee of unlearning, and in LoRA's low-rank setting it updates make redirection compete for capacity that may already be spent. Full-rollout training thus chases a moving late-position target while carrying ballast from earlier ones. We term this problem as "Stale Late-position Lock-in". This insight expects a training stability issues for full-rollout OPD – a prediction we later verify in §4.3

3 Our Method: Early Stopping Rollout (ESR) On Policy Distillation

3.1 Formulation

Let π_s denote the student and π_t the teacher. In standard on-policy reverse-KL distillation, the student generates a response $\mathbf{y} = (y_1, \ldots, y_T)$ conditioned on prompt x, and the loss is

$$\mathcal{L}_{\text{full}} = \mathbb{E}_{\mathbf{y}\sim\pi_s(\cdot|x)}\left[\sum_{t=1}^{T} \mathrm{KL}\big(\pi_s(\cdot \mid x, \mathbf{y}_{<t}) \parallel \pi_t(\cdot \mid x, \mathbf{y}_{<t})\big)\right]. \tag{1}$$

ESR (position cutoff N, with $N \ll T$ in practice) truncates the student rollout to its first N tokens, and the loss is computed over exactly those tokens:

$$\mathcal{L}_{\text{ESR}}(N) = \mathbb{E}_{\mathbf{y}\sim\pi_s(\cdot|x),\ |\mathbf{y}|\le N}\left[\sum_{t=1}^{\min(N,|\mathbf{y}|)} \mathrm{KL}\big(\pi_s(\cdot \mid x, \mathbf{y}_{<t}) \parallel \pi_t(\cdot \mid x, \mathbf{y}_{<t})\big)\right]. \tag{2}$$

If the student emits EOS before position N, the rollout terminates naturally. Everything else—generation temperature, LoRA target modules, optimizer, scorer—is unchanged from the standard on-policy KD loop.

4 Analysis on the Advancement of ESR

4.1 Setup

Models. We evaluate across multiple student-teacher pairs. Our primary student is Qwen2.5-Math-1.5B [Qwen Team, 2024], paired with teachers of increasing scale: Qwen3-1.7B, Qwen3-4B, and Qwen3-8B [Qwen Team, 2025]. We also tested Qwen2.5-Math-7B with Qwen3-14B to test student model size generality. To test cross-family generality, we also evaluate with gemma-2-2b-it (student) and gemma-3-4b-it (teacher).

Training. All experiments use LoRA [Hu et al., 2022] (r=32, α=64, all linear layers, lr = 5×10^{-5}) with reverse KL divergence loss. Temperature 0.7 for on-policy generation. Each training step processes a batch of 16 problems with 1 rollout per problem (n_{samples}=1, batch size 16). We train for 200 steps on all tasks, and saving checkpoints every 50 steps. Training data are drawn from from NuminaMath, CodeUltraFeedback, and glaive-function-calling-v2. Our method uses N=100 unless otherwise specified.

Evaluation. MATH-500 [Hendrycks et al., 2021, Lightman et al., 2023] with n=4 samples at temperature 0.7 (reporting avg@4), HumanEval/HumanEval+ [Chen et al., 2021, Austin et al., 2021, Liu et al., 2023] at temperature 0.0 (reporting pass@1), BFCL [Patil et al., 2025] with 600 problems reporting full accuracy: correct function name and arguments.

4.2 Overall performance

Table 1 presents the main results across tasks and model families. Table 2 presents ablation results with full finetuning. Table 2 shows model size scaling results. We report the best performance

Table 1: **Main results: ESR dominates Full Rollout across tasks and families.** Best score across training steps; metrics defined in §4 setup. ★: ESR surpasses the teacher reference. **Full Rollout** numbers mark the best step before degradation; $^{\downarrow}$ flags cells where Full Rollout further decays with training (more details in Appendix C).

Pair	**Method**	**MATH-500**		**Coding**		**BFCL**
		avg@4	pass@4	HE	HE+	full_acc
Qwen2.5-Math 1.5B → Qwen3 1.7B	Student baseline	50.95	72.80	31.10	26.20	2.70
	Teacher reference	65.30	77.00	39.60	36.00	54.00
	Full Rollout	62.35$^{\downarrow}$	74.60$^{\downarrow}$	40.20$^{\downarrow}$	35.40$^{\downarrow}$	58.20★
	ESR	**65.85** ★	**79.80** ★	**42.10** ★	**36.00**	**61.30** ★
Gemma-2 2B → Gemma-3 4B	Student baseline	13.45	28.20	23.78	20.10	73.17
	Teacher reference	66.60	74.80	20.70	20.10	72.83
	Full Rollout	22.95	31.40	22.00$^{\downarrow}$	17.70$^{\downarrow}$	76.83★
	ESR	**27.20**	**39.40**	**28.70** ★	**24.40** ★	**79.00** ★

Table 2: **FullFT and LoRA scaling on MATH-500. (Left)** FullFT for Qwen and Gemma student/teacher pairs. **(Right)** LoRA scaling across teacher and student sizes; $^{\downarrow}$ marks the **Full Rollout** peak before degradation. Both: best across training steps; **bold**: ESR beats **Full Rollout**.

Pair	**Method**	**avg@4**	**pass@4**
Qwen2.5-Math 1.5B → Qwen3 1.7B	student / teacher	50.95 / 65.30	72.80 / 77.00
	Full Rollout	58.20	75.40
	ESR	56.20	73.80
Gemma-2 2B → Gemma-3 4B	student / teacher	13.45 / 66.60	28.20 / 74.80
	Full Rollout	13.90	25.00
	ESR	**26.65**	**40.40**

Pair	**Method**	**avg@4**	**pass@4**
Qwen2.5-Math 1.5B → Qwen3 4B	student / teacher	50.95 / 77.95	72.80 / 86.40
	Full Rollout	67.45$^{\downarrow}$	80.60
	ESR	**68.95**	**81.00**
Qwen2.5-Math 1.5B → Qwen3 8B	student / teacher	50.95 / 77.70	72.80 / 87.80
	Full Rollout	66.75	81.60
	ESR	**67.85**	**82.20**
Qwen2.5-Math 7B → Qwen3 14B	student / teacher	53.60 / 76.15	75.00 / 83.20
	Full Rollout	68.85$^{\downarrow}$	80.00
	ESR	**68.95**	**81.20**

achieved across all training steps for each configuration. But notice that full rollout distillation frequently collapse to much lower performance as it trains. We denote the sessions that collapses with a down arrows.

ESRbeats full rollout across model families and tasks, and frequently surpasses teacher performance. Across the cells of Table 1, ESR beats full rollout on every cell, and in six cells ESR *surpasses the teacher* outright (Qwen math avg@4, pass@4; Qwen HumanEval and HumanEval+; Qwen BFCL full_acc; Gemma coding on all four columns; Gemma BFCL full_acc). The Gemma coding results have the biggest gap: full rollout drives HumanEval pass@1 to 22.00, *below* the 23.78 student baseline and worse than the 20.70 teacher, while ESR recovers 31.10. Qwen-math full rollout is marked $^{\downarrow}$ because its best number (62.35 avg@4) is a non-stable best step: under n=16 it collapses to 46.3% by step 100 (Appendix C.6).

ESRmatches or beats full rollout across full fine-tuning and LoRA settings. Table 2 reports FullFT on MATH-500 for both Qwen and Gemma. On Qwen **Full Rollout** 58.20 is slightly better than ours ESR 56.20 avg@4, but the gap is close. For Gemma ESR still dominates **Full Rollout** by $+12.75$pp avg@4 and $+15.40$pp pass@4. ESR is therefore the safer choice in both parameter regimes.

ESRscales across model sizes. The advantage persists as both teacher and student grow. Table 2 reports the results scaling both the teacher and the student. With the same student (Qwen2.5-Math-1.5B), we tried Qwen3-4B and 8B as the teacher. With Qwen3-4B, **Full Rollout** peaks at (67.45) and then *collapses* to 55.05 (Appendix C.6), while ESR stays stable and climbs to 68.95. Against the 8B teacher, ESR still edges **Full Rollout** (67.85 vs 66.75). The pattern persists at larger scale along with student model too: with a Qwen2.5-Math-7B student and Qwen3-14B teacher, **Full Rollout** again peaks at step 50 (68.85) and drifts to 62.40, while ESR holds stable across all checkpoints.

Figure 1: **(Left) Off-policy teacher drift.** The teacher loses accuracy quickly as the student-generated rollout grows over a few hundred tokens. MATH-500, avg@4 (n=4, t=0.7). Teacher = Qwen3-1.7B; student = Qwen2.5-Math-1.5B. After ∼300 student tokens the teacher has effectively been dragged down to student-baseline performance. **(Right) Rollout length N sweep on MATH-500.** LoRA, Qwen2.5-Math-1.5B → Qwen3-1.7B; best avg@4 across training steps. Full rollout and the undistilled baseline are shown as horizontal references. Performance saturates for $N \in [50, 200]$ and all beat full-seq.

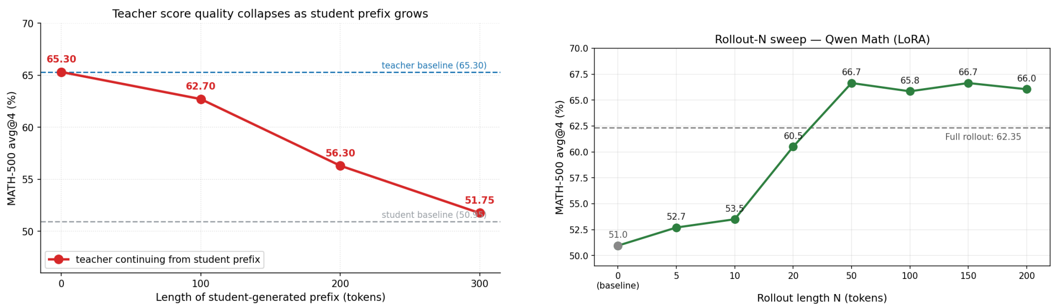


Table 3: **Training efficiency.** Single A6000 (48 GB), bs=16, student teacher Qwen3-1.7B. We report the average running time and GPU memory usage across student model generation, training and teacher scoring phases. Note that the real time usage can be larger if there isn't enough GPU to hold the student and teacher models together and requires model loading and unloading.

	Per-step wall time			*Peak GPU memory*		
Phase	**ESR (100 tok.)**	**Full rollout**	**Speedup**	**ESR (100 tok.)**	**Full rollout**	**Savings**
Teacher scoring	1 s	7 s	7×	7.3 GB	14.9 GB	2.0×
Student generation	5 s	180 s	36×	7.2 GB	8.9 GB	1.2×
Student training	2 s	7 s	3.5×	9.6 GB	39.5 GB	4.1×
Total (single step)	**8 s**	**194 s**	**24×**	**24.1 GB**	**63.3 GB**	**2.6×**
Total (200 steps)	**27 min**	**10.8 hr**	**24×**			

4.3 Stability of ESR

ESRis more robust than full rollout in training. Across our (pair × task × regime) matrix, full-rollout distillation degrades with training very frequently; ESR degrades nowhere. Appendix Table 12 reports each collapsing cell as a (peak → lowest) pair with the corresponding step indices. For example, Qwen-LoRA coding falls from 40.2 to 26.8; Qwen-LoRA math collapses from 65.60 at s50 to 43.80 ($\Delta = -21.80$pp, ending below the 50.95% student baseline); Gemma-LoRA collapse from 22.0 to 11.6 ($\Delta = -10.4$pp). Three of seven full-rollout collapses end below the student baseline. ESR generally hold a fluctuation within ±3pp of its peak across training (Tables 7, 9).

Early Stopping Rolloutis not sensitive to the choice of N. A natural question naturally occurs - how to choose where to stop? Is it sensitive? We conducted a set of sweeping experiments in Figure 1 sweeps N on MATH-500 and reveals a robust region: it reaches just as good performance starting from N=50 and remains stable to N=200. The method is not sensitive to the exact choice of N within a certain region.

4.4 Efficiency of ESR

Table 3 shows that ESR achieves a 24× wall-clock speedup and reduces peak training memory by $\sim 4\times$. The dominant cost in full-seq is autoregressive generation (180 s/step for sequences averaging ∼1000 tokens); ESR generates only N=100 tokens (5 s/step). Note that with ESR, all the student and teacher models can be put in one A6000 GPU comfortably. In our own practice, it saves a further big time overhead of model loading and unloading that we do not report here.

Figure 2: **(Left) Case study: early tokens encode strategy; later tokens mostly execute.** A representative MATH-500 trajectory. The first 100 tokens establish the geometric setup, name the unknown, and identify the key relationship (altitude bisects the leg)—these are the choices that determine whether the rollout will succeed. The last 100 tokens carry out an algebraic computation that any solver can finish once the strategy is fixed. ESR supervises the first window; the second follows for free. **(Right) Token selection strategies on MATH-500.** Qwen2.5-Math-1.5B $\rightarrow$ Qwen3-1.7B, LoRA, n_{samples}=1, best avg@4 across training steps. All selectors operate on full-length rollouts; ESR is the only one that additionally truncates the rollout. Baseline (no distillation) 50.95%; full-seq 62.35%. H_s: student entropy; H_t: teacher entropy; RKL: reverse KL.

Problem. In an isosceles right triangle, the altitude to the hypotenuse has length $4\sqrt{2}$. What is the area?

First 100 tokens (strategy). *"Let's start by understanding the problem. In an isosceles right triangle, the two legs are equal, and the hypotenuse is $\sqrt{2}$ times the leg. If we denote each leg by a, then hypotenuse $= a\sqrt{2}$. The altitude to the hypotenuse is half the leg, because the altitude bisects the..."*

Last 100 tokens (execution). *"... altitude from the right-angle vertex to the hypotenuse. This altitude has length $h = \frac{a \cdot a}{a\sqrt{2}} = \frac{a^2}{a\sqrt{2}} = \frac{a}{\sqrt{2}} = \frac{a\sqrt{2}}{2}$. Setting this equal to $4\sqrt{2}$: $\frac{a\sqrt{2}}{2} = 4\sqrt{2}$, so $a\sqrt{2} = 8\sqrt{2}$, giving $a = 8$. The area is $\frac{1}{2} \times 8 \times 8 = \boxed{32}$."*

Selection method	k	Best avg@4
ESR-100 (ours)	100	**65.85**
Full-seq (all tokens)	—	62.35
Top-RKL	100	53.35
Top-H_t (teacher entropy)	100	63.30
Top-H_s (student entropy)	100	62.70
Top-H_s	200	53.20
RKL$\cdot H_s$	100	56.90
$H_t \cdot H_s$ (product)	100	55.35
RKL$\cdot H_t \cdot H_s$ (triple product)	100	57.90

5 Additional Reasons Why Does ESR Work?

The surprising effectiveness of Early Stopping Rollout probes us to investigate deeper beyond what we have identified Section 2. We structure this section around three questions: **(§5.1)** Is the early-position effect induced by by high-KL or high-entropy tokens? **(§5.2)** If not, what additional structural reasons make the early window privileged? **(§5.3)** Why does ESR sometimes *exceed* the teacher?

5.1 What Does The Effectiveness of ESR Stem From?

In Section 2, we identified the "Off-policy Teacher Drift" and "Stale Late-position Lock-in" problem that can cause the late-position tokens to provide degrading signals. During the experiment analysis, we find that, as shown in Figure 3, early positions simultaneously have high KL divergence between student and teacher model, and high token entropy from both student and teacher models. This finding probes us to wonder if the effectiveness is intrinsically induced by the KL and entropy. To control these factors, we conduct a series of ablation experiments: pick the same amount (100) of tokens based on the highest KL divergence, highest student/teacher entropy or with them in combination with ESR, regardless of position (Figure 2). If the effectiveness is indeed induced by the KL or entropy, then they should reproduce the same or better results.

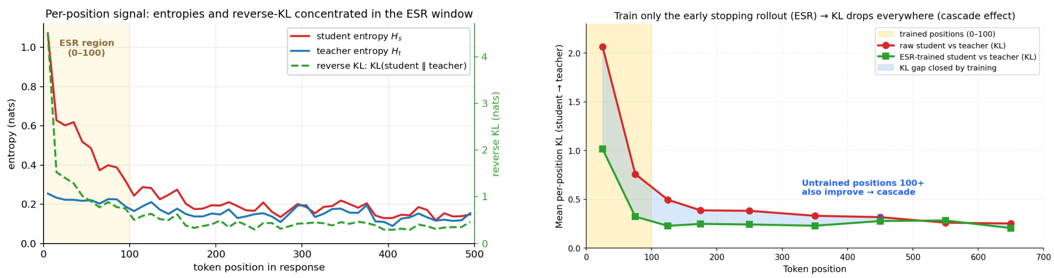
To our surprise, all underperform ESR, and most of them also much underperform the full rollout results. Teacher or student entropy based selection can match the full sequence by them alone, but their combination falls short significantly. What's also interesting is that KL divergence measure, the direct calculation of the loss magnitude, barely works. It only improves the baseline (50.95%) for about 3 percent. And more surprisingly, we find that the largest 100 tokens of KL occupies around 93% of the entire trajectory loss. This shows that the tokens that has larger signals are not necessarily the ones that have effective signals.

Therefore, althought we still don't exclude KL and entropy as potential mediator factor, we exclude them to be the sole factors that causes early tokens to be special. Position, therefore, should be considered an independent token selection dimension for the future.

5.2 The Convergence Cascade Effect of ESR

We find an additional important mechanism that assists in ESR's effectiveness. Even we only trained the first N tokens with ESR, training, per-position KL divergence beyond $[0, N]$ region still drops by 30–40% (Figure 3). This shows that the student can pick up the teacher's "global mindset" even with just the beginning tokens. One reason that we suspect is that the beginning tokens often consist of

Figure 3: **(Left)** Early tokens are high on student entropy, teacher entropy, and KL divergence simultaneously. **(Right) The convergence cascade.** Per-position KL between distilled student and teacher, before vs. after ESRtraining. Yellow band: positions $[0, 100]$ that actually receive training loss. Blue band: the KL gap closed by training. Positions $100+$—which see no direct training signal—drop to the same KL as the trained region, confirming that alignment on the early window cascades through the autoregressive rollout.



problem framing and strategic planning content. Therefore once the student picks up how to frame problems and plan the strategy, the later content naturally follows.

Moreover, recently Cloud et al. [2025] shows student models may be able to learn the teacher's deep internal preference even with random numbers generated by the teacher, called "subliminal learning". Therefore, the early tokens may suffice to inject the subliminal mindset to the student.

5.3 Why Does ESR Even Exceed the Teacher?

ESR exceeds the teacher in most of the main experiments1. Even the full rollout model slightly exceeds the teacher in function calling experiments, although weaker than our method. This shows that student has the potential exceed the teacher even in normal OPD, and our method amplifies it. Why is so? Isn't teacher supposed to be the upper bound?

We propose the reason lies in the mechanism of reverse KL $\mathrm{KL}(\pi_s \,\|\, \pi_t)$, which has a mode-seeking behavior: it penalizes the student for putting mass on tokens the teacher does not support, but not for concentrating mass on a single supported token. Therefore, the student has the possibility to land on a submode of the teacher that is actually better. We visualize this mechanism in figure 4.

To verify it is the case, we first scan through the behaviors difference of the distilled models. Surprisingly, models trained with our method produces 2x to 3x shorter sequences than not only the teacher and the full rollout versions, and even shorter than itself. Note that the teacher model is a more verbose model than the student. So learning from such teacher generally leads to longer sequences. The full rollout version is indeed the case. It produces sequence even much longer than the teacher. The juxtaposing direction learned from the same teacher caused just by the rollout length shows how important this choice leads to. Without the impact of the late-position tokens, the model preserves its original succint style and learns the teacher's strategy altogether, leading to a more desirable outcome than simply copying the teacher.

Furthermore, we how student's probability output mode aligns with the teacher quantitatively. We take out the top 10 % of highest KL tokens after the training that can tell the model's behavior difference most saliently. We check the frequency when the student's top choice agrees with the teacher's top choice, top 2-5 choices, and beyond. We find that, indeed, our method produces a model that is more commited to top 2-5 choices of the teacher rather than the first choice. And the top 1 choice average probability of our method is higher than that of the full rollout, showing our method leads the model to be more confident in its choice. This exactly shows that ESRsteers the model to commit to a secondary mode in the teacher model.

Behavioral signature. The hypothesis predicts that ESR-trained students should (a) produce shorter, less-hedging rollouts than the teacher, and (b) commit hard, but on *non-argmax* teacher modes. Table 4 confirms both. ESR-100 rollouts are shorter than the base student, the teacher, *and* full-seq across the entire length distribution — even the top-decile ($\sim$800 tokens) is below the base student's median ($\sim$860). On the top-10% highest-KL positions (n=110,894), ESR raises student

Figure 4: **Mode-seeking on the planning region.** Schematic of reverse-KL behaviour at a multi-modal planning position. The teacher supports two modes (e.g. a verbose plan and a concise correct plan); reverse KL penalises student mass outside the support but not concentration within it. A short early-window loss can collapse the student onto the better supported mode, allowing the student to exceed the teacher's average behaviour. Full-sequence training reverts the student toward the averaged teacher across late positions, undoing the concentration.

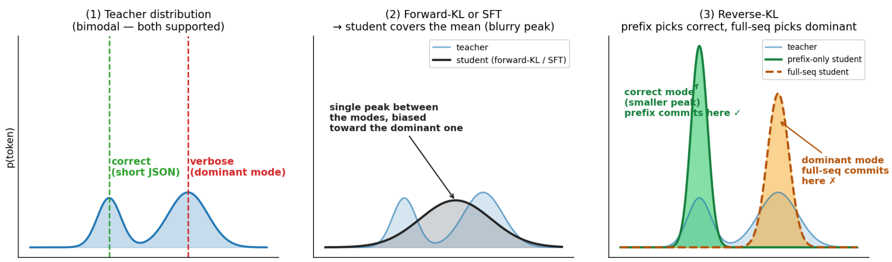


Table 4: **Behavioral Modes Analysis.** *Left:* Response length distribution on MATH-500. ESRlearns an output style that dramatically shorter than full rollout, teacher, and even its own baseline model. We show the length of the 10 percentile, 50 percentile (media), 90 percentile and the average length. *Right:* How student's output aligns with the teacher's distribution modes. "Top-1 Prob" shows the probability of the top 1 token's probability mass, indicating how much the student commits in its first choice. "=tch. top-1" means the percentage of times where the student's first choice is same as that of the teacher. "$\in$ tch. top 2–5" means the percentage of times where the student's first choice is the teachers 2nd to 5th choice. And "$\notin$ tch. top-5" are the times the student's first choice isn't in the teacher's top 5 list. Our method tends to choose teacher's non-first choice and is more confident in its choice than the full rollout version.

	Response length (tokens)					Student commitment & mode alignment			
Model	**10th pctile**	**50th pctile**	**90th pctile**	**Mean**	**Ratio**	**Top-1 prob**	**= tch. top-1**	**$\in$ tch. top 2–5**	**$\notin$ tch. top-5**
Base student	∼400	∼860	∼1,500	∼990	2.2×	0.71	28.6%	59.8%	11.5%
Teacher (Q3-1.7B)	∼700	∼1,150	∼1,800	∼1,210	2.6×	—	—	—	—
Full Rollout	∼1,190	∼1,530	∼1,770	∼1,480	3.2×	0.77	**45.7%**	44.6%	9.7%
ESR	∼190	∼380	∼800	∼460	1.0×	**0.79**	41.9%	**47.4%**	10.7%

top-1 probability to 0.79 (vs 0.77 full-seq, 0.71 base) while *lowering* agreement with the teacher's argmax to 41.9% (vs 45.7% full-seq) and raising the rate at which the student picks tokens in the teacher's top-2 to top-5 to 47.4% (vs 44.6%). ESR commits harder, but on different teacher-supported tokens; full-seq commits toward the teacher's argmax and inherits the teacher's mistakes on those positions.

This is the mode-seeking mechanism in numbers. When one of those teacher-supported alternatives happens to be the correct mode, ESR exceeds the teacher; full-seq cannot, because supervising the late tail pulls the student toward the teacher's argmax and the teacher's mistakes along with it. A full proof of this hypothesis is out of scope; we note only that it is the natural reverse-KL-mode-seeking story applied to the planning region, and that it is consistent with every surpass-teacher cell we observe.

6 Related Work

Knowledge Distillation for Language Models. Knowledge distillation [Hinton et al., 2015, Kim and Rush, 2016] for autoregressive LLMs hinges on the choice of divergence and data distribution. Gu et al. [2024] argued for reverse KL, and Agarwal et al. [2024] introduced on-policy GKD, which supervises student rollouts with teacher logits. Subsequent work has explored skew KL with adaptive off-policy schedules [Ko et al., 2024], generalized f-divergences [Wen et al., 2023], mode- vs. mean-seeking dynamics [Wu et al., 2025], and speculative-decoding-style filtering [Xu et al., 2025].

We build on the on-policy reverse-KL paradigm but ask an orthogonal question: holding divergence and data distribution fixed, *which token positions along the student's rollout carry useful signal?*

Token-Level Importance in Distillation and Reasoning. Not all tokens contribute equally to learning. Wang et al. [2025] showed that $\sim$20% high-entropy "forking tokens" suffice for RL gradients; Vassoyan et al. [2025] relaxed KL on critical tokens to enable exploration; Li et al. [2025] identified a preplan-and-anchor mechanism in attention; and Singh and Hakkani-Tür [2026] pruned reasoning chains by functional importance. In distillation specifically, recent work selects tokens via teacher verification [Huang et al., 2025], adaptive temperature [Xie et al., 2026], student entropy along position/class/sample axes [Tavor et al., 2026], and entropy plus preference ranking [Kim and Baek, 2026]. Our Section 5.1 ablation contradicts the shared premise: selecting tokens by top-KL, various top-entropy variants, soft-weighted joint entropy, or top-K-renorm RKL[1] all underperform ESR and most underperform plain full-sequence training. Position is the load-bearing axis; scalar saliencies are not. This also reconciles Wang et al. [2025] with our findings: in on-policy distillation, high-entropy tokens are disproportionately concentrated in the uncontaminated early window.

Training Stability and Sequence Length. Full-sequence on-policy distillation is unstable: likelihood training degenerates into bland, repetitive text [Holtzman et al., 2020], and self-training drives model collapse [Shumailov et al., 2024]. Jang et al. [2026] proposed a geometric logit-space bridge to suppress harmful gradients on low-confidence tokens. On the length axis, Chen et al. [2025] showed that supervised training on the first 50% of tokens retains $\sim$91% of full-sequence performance, and Yan et al. [2026] addressed exposure bias in long-CoT distillation via temperature scheduling. Our approach is complementary: restricting the loss scope to early positions, where teacher–student agreement is naturally highest, yields stability through a simpler mechanism and eliminates the degenerate repetition unique to on-policy training.

Curriculum Learning. Curriculum learning [Bengio et al., 2009] and scheduled sampling [Bengio et al., 2015] gradually increase task difficulty. For LLM pretraining, sequence-length warmup stabilizes training [Li et al., 2022] and variable-length curricula accelerate convergence [Pouransari et al., 2024]. Our progressive position schedule applies the curriculum principle to the distillation loss scope rather than to the input length—a form of curriculum specific to knowledge distillation.

Reasoning Distillation. Chain-of-thought prompting [Wei et al., 2022] made intermediate reasoning the unit of transfer, and DeepSeek-R1 [DeepSeek-AI, 2025] showed that SFT-distilling a strong reasoner into a smaller model is often competitive with direct RL. On reasoning tasks (MATH-500, HumanEval/MBPP, BFCL) and a 1.5B-math-student $\rightarrow$ 8B-teacher sweep, ESR surpasses the teacher on several cells, indicating that gains beyond the teacher are possible when the mode-seeking reverse-KL loss is restricted to the planning region—explaining why the on-policy, early-stopped-rollout regime is more efficient and stable than full-sequence on-policy distillation, which frequently degrades on coding and math.

7 Conclusion

In on-policy knowledge distillation, the teacher's per-token supervision quality is a strong, monotonic function of position: early tokens carry strong, independent signal about reasoning strategy; late tokens, scored on the student's own contaminated previous output, contribute weak, dependent signal that is actively harmful when trained on. A one-line intervention — restrict the loss and the rollout to the first N response tokens — yields four simultaneous wins: (i) **better**, matching or exceeding full-sequence distillation on every task/family/regime we test and surpassing the teacher in nine cells; (ii) **faster**, 20–35$\times$ wall-clock with 4$\times$ less GPU memory; (iii) **stabler**, eliminating HumanEval collapse, BFCL parse-rate collapse, and the MATH-500 boxed-repetition failure mode; and (iv) **structural**, in that selecting the same number of tokens by any KL or entropy saliency fails — position is an independent axis, not a proxy for scalar surprise. Teaching the student how to begin is sufficient to improve the entire response.

Limitations. Our experiments are conducted on small open-source models with a limited data budget; whether the ESR story holds at industrial scale (much larger students, teachers, and training

[1] TODO: resolve and cite the top-K-renorm RKL method (Fu et al., 2025).

corpora) remains unclear. We also do not test multi-modality or long-horizon tasks, both of which may exhibit different positional signal-quality patterns.

Broader Impacts. ESR reduces the wall-clock and memory cost of on-policy distillation by 20–35$\times$ and $\sim 4\times$ respectively, which lowers the barrier to legitimate distillation research for academic and small-industry groups that cannot afford full-rollout pipelines. The same efficiency gain has a dual-use aspect: it lowers the cost of distilling capabilities from undisclosed or proprietary teacher models, with potential implications for model-extraction and IP. We use only publicly released open-weight teachers and standard public benchmarks in this work, but practitioners deploying ESR on closed teachers should respect the teachers' terms of use.

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A Training Efficiency

Table 5: **Training efficiency, detailed breakdown.** Per-step wall-clock time on a single A6000 (48GB). Student: Qwen2.5-Math-1.5B, LoRA, bs=16.

Teacher	Method	Gen (s)	Score (s)	Train (s)	Total (s)
Q3-1.7B	Full-seq	100–731	3–10	5–12	$\sim$280
	Ours (N=100)	$\sim$5	$\sim$1	$\sim$2	$\sim$8
Q3-4B	Full-seq	97–733	3–9	6–10	$\sim$210
	Ours (N=100)	$\sim$5	$\sim$1	$\sim$2	$\sim$8
Q3-8B	Full-seq	96–230	7–11	1–6	$\sim$170[†]
	Ours (N=100)	$\sim$5	$\sim$1	$\sim$2	$\sim$8

[†] Frequent OOMs; 48GB insufficient for 8B teacher + vLLM + student.

B Additional Student-Teacher Pairs

Table 6: **Additional student-teacher pairs.** All use ESR-100, LoRA, 3,200 problems.

Student	Teacher	MATH-500 avg@4	BFCL full_acc	HumanEval pass@1
Qwen3-1.7B	— (baseline)	69.20%	59.80%	39.63%
Qwen3-1.7B	Qwen3-4B	69.20%	73.70%	44.51%
Qwen3-1.7B	Qwen3-8B	69.15%	74.00%	40.24%
Qwen3-4B	— (baseline)	77.95%	73.00%	73.17%
Qwen3-4B	Qwen3-8B	77.05%	76.80%	71.34%

C Full Experimental Results

C.1 Math Results: Per-Step Performance

Table 7 presents per-step results for the primary math experiments (LoRA, n=1, 3,200 problems).

Table 7: **MATH-500 per-step results.** LoRA, n=1, bs=16, 3,200 problems. Baseline: 50.95% avg@4.

Method	Metric	Step 50	Step 100	Step 150	Step 200
ESR-50	avg@4	62.35	66.05	**66.65**	64.85
	maj@4	69.40	72.00	71.00	71.20
	pass@4	77.20	79.40	81.00	79.60
ESR-100	avg@4	63.75	64.45	65.15	**65.85**
	maj@4	70.00	68.40	69.60	70.80
	pass@4	79.80	78.40	80.20	79.80
ESR-150	avg@4	65.35	**66.65**	65.30	65.75
	maj@4	66.80	67.00	66.30	67.30
	pass@4	79.00	81.00	78.20	80.00
ESR-200	avg@4	**66.05**	64.65	65.10	65.55
	maj@4	71.20	68.40	70.00	71.20
	pass@4	81.00	79.80	80.60	80.60

Table 8: **Complete MATH-500 results, LoRA, n=16, bs=16.** Best per configuration in **bold**.

Config	Step 50	Step 100	Step 150	Step 200
avg@4				
ESR-5	55.15	55.65	**56.50**	56.30
ESR-10	**59.50**	56.40	56.30	56.00
ESR-20	**60.20**	57.35	56.40	56.55
ESR-50	**62.45**	62.15	60.50	61.90
ESR-100	61.15	63.55	**64.25**	63.85
ESR-150	63.55	**65.70**	64.80	64.40
ESR-200	63.25	**66.75**	64.55	65.25
Progressive	61.10	62.05	61.95	**62.30**
Full-seq†	**65.60**	46.30	49.70	43.80
Full-seq (first-boxed)	**65.55**	64.95	65.55	64.95
pass@4				
ESR-5	74.0	74.8	**76.0**	73.4
ESR-10	**75.8**	73.2	75.4	74.2
ESR-20	**77.0**	76.0	73.4	75.8
ESR-50	77.0	78.4	77.6	**78.6**
ESR-100	77.6	79.2	**80.2**	80.2
ESR-150	79.6	77.4	**80.0**	79.4
ESR-200	79.6	**80.2**	79.4	79.2
Progressive	77.2	77.8	78.2	**78.8**
Full-seq	**80.0**	72.2	77.0	74.2

† Last-boxed extraction; degrades due to repetition.

Table 9: **Complete coding results, LoRA.** HumanEval (HE) and MBPP pass@1.

Config	Metric	s50	s100	s150	s200	s250	s300	s350	s400
ESR-50	HE	37.8	39.0	39.6	41.5	40.2	40.9	**42.1**	40.9
	MBPP	52.1	48.9	46.0	46.0	47.6	46.3	46.6	46.3
ESR-100	HE	37.2	39.0	**42.1**	37.8	39.0	37.8	37.8	38.4
	MBPP	52.1	49.7	49.2	49.2	48.4	49.2	47.9	48.9
ESR-150	HE	36.6	35.4	36.6	39.0	**41.5**	39.6	38.4	37.2
	MBPP	51.3	50.0	48.4	50.5	49.5	50.3	49.7	48.7
Full-seq	HE	**40.2**	31.7	32.3	32.9	27.4	28.0	26.8	26.8
	MBPP	52.6	48.9	49.5	48.1	47.9	47.1	48.4	47.6

C.2 Math Results: Earlier Series (n=16, 200 problems)

C.3 Coding Results

C.4 Function Calling Results

Table 10: **Function calling results (BFCL), LoRA.** Name accuracy / Full accuracy / Parse rate. Best full_acc in **bold**.

Method	Best Step	Name Acc	Full Acc	Parse Rate
Baseline	—	9.70%	2.70%	24.20%
Teacher (Qwen3-1.7B)	—	75.30%	54.00%	75.30%
ESR-50	200	95.20%	57.20%	98.30%
ESR-100	**100**	86.20%	**61.30%**	91.30%
ESR-150	200	88.70%	61.50%	92.50%
ESR-200	200	80.80%	54.50%	90.20%
Full-seq	100	81.00%	58.20%	86.70%

C.5 Scaling Experiment Results

Table 11: **Scaling results.** All use ESR-100, LoRA, 3,200 problems. Best across training steps. Baseline performance shown for reference.

Student → Teacher	Math avg@4	BFCL full_acc	HE pass@1	MBPP+ pass@1
M-1.5B baseline	50.95%	2.70%	—	—
M-1.5B → Q3-1.7B	65.85%	61.30%	42.07%	44.44%
M-1.5B → Q3-4B	68.95%	45.80%	43.29%	47.62%
M-1.5B → Q3-8B	67.85%	57.00%	41.46%	44.44%
Q3-1.7B baseline	69.20%	59.80%	39.63%	44.71%
Q3-1.7B → Q3-4B	69.20%	73.70%	44.51%	47.62%
Q3-1.7B → Q3-8B	69.15%	74.00%	40.24%	48.15%
Q3-4B baseline	77.95%	73.00%	73.17%	55.56%
Q3-4B → Q3-8B	77.05%	76.80%	71.34%	56.35%

C.6 Full-Sequence Degradation Details

Table 12 summarises every full-rollout cell (across pairs, tasks, and regimes) for which we have a per-step trajectory on disk and observe degradation across training. For each cell we report the *peak* score (best checkpoint across training) and the *lowest* score (worst checkpoint after the peak), together with the step indices at which they occur. Bold Δ values indicate cells where Full Rollout drops below the student baseline. The Qwen-math n=16 row (boxed-repetition collapse) is documented in greater detail in Table 13 below.

Per-step trajectories for the seven cells in Table 12:

- **Qwen-1.5B→Q3-1.7B, LoRA, HumanEval (pass@1):** 40.2 (s50) → 31.7 → 32.3 → 32.9 → 27.4 → 28.0 → 26.8 → 26.8 (s400).
- **Qwen-1.5B→Q3-1.7B, LoRA, HumanEval+ (pass@1):** 35.4 (s50) → 27.4 → 29.9 → 29.9 → 25.0 → 26.2 → 25.0 → 25.0 (s400).
- **Qwen-1.5B→Q3-1.7B, FullFT, HumanEval (pass@1):** 32.3 → 32.3 → 36.6 (s150) → 36.0 → 31.1 → 30.5 → 31.7 → 31.1 (s400).
- **Qwen-1.5B→Q3-1.7B, FullFT, HumanEval+ (pass@1):** 27.4 → 28.7 → 31.7 (s150) → 30.5 → 26.2 → 26.2 → 26.8 → 26.2 (s400).
- **Gemma-2-2B→Gemma-3-4B, LoRA, HumanEval (pass@1):** 22.0 (s50) → 14.6 → 11.6 → 11.6 (s200). Falls below the 23.78 student baseline at s100 and stays there.

Table 12: **Full-rollout collapses across training (peak → lowest).** Each row is a single Full Rollout experiment; we report the highest and lowest checkpoint scores across all saved steps. "Δ" is lowest − peak. Cells with † end the run *below the student baseline*. Trajectories are taken from the per-step eval logs on the training cluster (`docs/main_results.md`, `docs/scaling_results.md`). ESR on the same cells (not shown here) stays within ±1pp of its peak across training; see Tables 7 and 9.

Pair	Regime	Task	Metric	Peak (step)	Lowest (step)	Δ
Qwen-1.5B → Q3-1.7B	LoRA	MATH-500 (n=16)	avg@4	65.60 (s50)	43.80 (s200)†	**−21.80**
Qwen-1.5B → Q3-4B	LoRA	MATH-500	avg@4	67.45 (s50)	54.45 (s100)	−13.00
Qwen-1.5B → Q3-4B	LoRA	MATH-500	maj@4	73.20 (s50)	63.40 (s200)	−9.80
Qwen-1.5B → Q3-1.7B	LoRA	HumanEval	pass@1	40.2 (s50)	26.8 (s350)	−13.4
Qwen-1.5B → Q3-1.7B	LoRA	HumanEval+	pass@1	35.4 (s50)	25.0 (s250)	−10.4
Qwen-1.5B → Q3-1.7B	FullFT	HumanEval	pass@1	36.6 (s150)	30.5 (s300)	−6.1
Qwen-1.5B → Q3-1.7B	FullFT	HumanEval+	pass@1	31.7 (s150)	26.2 (s250)	−5.5
Gemma-2-2B → Gemma-3-4B	LoRA	HumanEval	pass@1	22.0 (s50)	11.6 (s150)†	**−10.4**

† Lowest score is below the student baseline (Qwen-Math-1.5B MATH-500: 50.95%; Gemma-2-2B HumanEval: 23.78%).

- **Qwen-1.5B→Q3-4B, LoRA, MATH-500 avg@4:** 67.45 (s50) → 54.45 → 58.85 → 55.05 (s200) (boxed-repetition pattern; ESR-100 on the same pair climbs to 68.95).
- **Qwen-1.5B→Q3-1.7B, LoRA, MATH-500** (n=16) **avg@4:** 65.60 (s50) → 46.30 → 49.70 → 43.80 (s200). Boxed-repetition collapse; statistics in Table 13.

Table 13: **Boxed repetition progression** in full-sequence LoRA distillation on MATH-500 (n=16, bs=16). Companion to the corresponding row of Table 12.

Statistic	Step 50	Step 100	Step 150	Step 200
Multi-boxed responses	2.9%	91.5%	80.5%	89.1%
Avg `\boxed{}` per response	1.0	88.1	54.1	58.0
Repetitive responses	5.9%	78.8%	58.3%	77.5%
Avg response length (chars)	1,388	4,023	3,585	4,558
Official avg@4	65.60%	46.30%	49.70%	43.80%
Relaxed avg@4 (GT in text)	74.60%	70.90%	71.90%	70.30%

D KL-Profile Diagnostic for Choosing N

A practical question for a new student–teacher pair is: what value of N should I use? Our position-sweep (Section 4.3) shows a wide plateau on Qwen math ($N \in [50, 200]$, avg@4 within ∼1pp); the same plateau appears on Gemma math ($N \approx 100$). These plateau locations coincide with the position at which the cumulative per-position KL $\sum_{k \leq N} \mathbb{E}[\mathrm{KL}_k]$ reaches ∼45% of the full-trajectory total, measured on a small (∼1k trajectory) pre-training rollout. This suggests a cheap rule of thumb: run a 1k-trajectory per-position KL scan before training, and pick N at the 45% cumulative-KL point. We verify this on two cells (Qwen, Gemma math) and leave a full six-cell verification ({Qwen, Gemma} × {math, coding, funcall}) as a short follow-up ([TBD: KL-profile scan]); the diagnostic is reported here as a practical recipe rather than a claim.

E Token Classification Methodology

We classify each token into six categories based on string matching:

1. **planning**: Reasoning keywords ("To", "Let", "First", "Step", "We", "Given", "Therefore", "Thus", "Since").
2. **structural**: Punctuation, whitespace, formatting tokens.
3. **math_number**: Digits (0–9).
4. **math_operator**: Arithmetic operators ($+$, $-$, $\times$, $/$, $=$).

5. **math_latex**: LaTeX delimiters (`\(`, `\[`).
6. **continuation**: All others.

Table 14: **Mean KL by token category and position range.**

Category	0–4	5–19	20–49	50–99	100–199	200–499
planning	4.50	0.79	1.49	1.66	1.49	2.37
structural	3.26	1.46	1.60	0.93	0.60	0.86
math_number	1.49	0.60	0.74	0.28	0.17	0.13
math_operator	7.30	1.84	0.81	0.37	0.21	0.14
math_latex	8.84	9.23	6.50	4.95	2.97	1.87
continuation	1.94	1.12	1.19	0.89	0.70	0.48

Table 15: **Top 20 highest-KL tokens** (minimum 50 occurrences across 10,000 trajectories).

Rank	Token	Category	Count	Mean KL
1	"Solution"	planning	152	21.93
2	"Analysis"	continuation	125	16.49
3	`\[`	math_latex	7,152	13.21
4	"examines"	continuation	74	11.51
5	"He"	continuation	150	10.80
6	`\(`	math_latex	21,243	10.30
7	"First"	planning	1,706	9.98
8	"tests"	continuation	52	8.71
9	`\\`	math_latex	82	8.68
10	"There"	continuation	201	8.28
11	"Therefore"	planning	4,913	7.95
12	"Identify"	continuation	1,345	7.78
13	"To"	planning	8,806	6.90
14	"When"	continuation	89	6.62
15	"This"	continuation	621	6.34
16	"Thus"	planning	2,547	6.24
17	"First" (space)	planning	259	6.10
18	"To" (space)	planning	1,174	5.37
19	"The"	planning	2,626	5.25
20	"Next"	planning	1,753	5.03

F Assets and Licenses

Table 16 lists the models and datasets used in this paper, with their providers and licenses. All assets are used in accordance with their respective terms of use.

Table 16: **Assets used in this paper.** All licenses verified at time of submission.

Asset	Type	Provider	License
Qwen2.5-Math-1.5B	Model	Alibaba	Apache 2.0
Qwen2.5-Math-7B	Model	Alibaba	Apache 2.0
Qwen3-1.7B / 4B / 8B / 14B	Model	Alibaba	Apache 2.0
Gemma-2-2B	Model	Google	Gemma Terms of Use
Gemma-3-4B	Model	Google	Gemma Terms of Use
NuminaMath	Dataset	Numina	Apache 2.0
CodeUltraFeedback	Dataset	Coseal	MIT
glaive-function-calling-v2	Dataset	Glaive AI	Apache 2.0
MATH-500	Benchmark	Hendrycks et al.	MIT
HumanEval / HumanEval+	Benchmark	OpenAI / EvalPlus	MIT
BFCL	Benchmark	UC Berkeley	Apache 2.0

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   Answer: [Yes]

   Justification: The abstract and §1 state that ESR matches or exceeds full-rollout distillation while being 20–35$\times$ faster with $\sim 4\times$ less memory; each claim maps to Table 1, §4.3, §4.4, and §5.1.

2. **Limitations**

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   Justification: §4 lists student/teacher pairs, datasets, all hyperparameters, the loss, and evaluation protocols; Appendices A and C give per-step timing and results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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   Justification: We will release the training code and LoRA adapters publicly after acceptance; the methodology in §4 is detailed enough to reproduce the main results in the meantime.

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   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: Reported numbers are single-seed best-step results; robustness comes from breadth across families, tasks, and teacher scales rather than per-cell variance, and this is acknowledged in §7.

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   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix A (Table 5) reports per-step wall-clock on a single A6000 (48GB) for each teacher scale, broken down by phase.

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   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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